

Mastering Flexbox: A Complete Guide

Understanding Flexbox

Flexbox (Flexible Box Layout) is a CSS layout module designed to distribute space and align items efficiently, especially in dynamic or unknown sizes.

Key Flexbox Properties

1. Parent (Flex Container) Properties

- **display:** flex | inline-flex
 - Defines a flex container. Example:

```
.container {  
  display: flex;  
}
```

- **flex-direction:** row | row-reverse | column | column-reverse
Specifies the direction of the main axis. Example:

```
.container {  
  flex-direction: column;  
}
```

- **justify-content:** flex-start | flex-end | center | space-between | space-around | space-evenly
 - Aligns items along the main axis. Example:

```
.container {  
  justify-content: space-between;  
}
```

- **align-items:** flex-start | flex-end | center | baseline | stretch
 - Aligns items along the cross axis. Example:

```
.container {  
    align-items: center;  
}
```

- **align-content:** flex-start | flex-end | center | space-between | space-around | stretch
 - Aligns multiple flex lines.
- **flex-wrap:** nowrap | wrap | wrap-reverse
 - Defines whether items should wrap.
- **gap, row-gap, column-gap:**
 - Controls spacing between flex items.

2. Child (Flex Item) Properties

- **flex:** flex-grow flex-shrink flex-basis
 - A shorthand for the three properties below. Example:

```
.item {  
    flex: 1 1 100px;  
}
```
- **flex-grow:** A number defining how much an item should grow relative to others.
- **flex-shrink:** A number defining how much an item should shrink relative to others.
- **flex-basis:** Defines the initial size of a flex item before distribution.
- **align-self:** auto | flex-start | flex-end | center | baseline | stretch
 - Allows individual items to override align-items.

- **order:** A number defining the order of the item.